# Activity 2.2: Compound Inequalities & Writing Inequalities

**A**

Name: _____ Partner's name: _____ Date: _____

**Directions:** Solve each compound inequality (#1–5 and #8), or write the inequality (#6–7). Write answers using inequality symbols.

*Work on your own. Then compare answers with your partner after each problem. If they don't match, find the mistake together before moving on.*

**1**

$2 < x + 1 < 6$

**Answer:** _____ Match? ☐ Yes ☐ No

**2**

$-4 \leq 2x \leq 8$

**Answer:** _____ Match? ☐ Yes ☐ No

**3**

$1 \leq 3x - 2 < 10$

**Answer:** _____ Match? ☐ Yes ☐ No

**4**

$x - 3 < -2$ or $x - 3 > 4$

**Answer:** _____ Match? ☐ Yes ☐ No

**5**

$3x \geq 12$ or $x + 5 < 2$

**Answer:** _____ Match? ☐ Yes ☐ No

**6**

A number $n$ is greater than or equal to 7 but less than 12. Write a compound inequality.

**Answer:** _____ Match? ☐ Yes ☐ No

**7**

You have \$50 to spend. A hoodie costs \$18 and each shirt costs \$8. Write and solve an inequality for the number of shirts, $s$, you can buy.

**Answer:** _____ Match? ☐ Yes ☐ No

**8**

$-3 < -x \leq 2$

**Answer:** _____ Match? ☐ Yes ☐ No

# Activity 2.2: Compound Inequalities & Writing Inequalities

**B**

Name: _____ Partner's name: _____ Date: _____

**Directions:** Solve each compound inequality (#1–5 and #8), or write the inequality (#6–7). Write answers using inequality symbols.

*Work on your own. Then compare answers with your partner after each problem. If they don't match, find the mistake together before moving on.*

**1**

$3 < x + 2 < 7$

**Answer:** _____ Match? ☐ Yes ☐ No

**2**

$-6 \leq 3x \leq 12$

**Answer:** _____ Match? ☐ Yes ☐ No

**3**

$3 \leq 2x + 1 < 9$

**Answer:** _____ Match? ☐ Yes ☐ No

**4**

$x + 2 < 3 \quad \text{or} \quad x + 2 > 9$

**Answer:** _____ Match? ☐ Yes ☐ No

**5**

$2x \geq 8 \quad \text{or} \quad x + 4 < 1$

**Answer:** _____ Match? ☐ Yes ☐ No

**6**

A number $n$ is at least 7 and also less than 12. Write a compound inequality.

**Answer:** _____ Match? ☐ Yes ☐ No

**7**

You have \$60 to spend. A backpack costs \$28 and each notebook costs \$8. Write and solve an inequality for the number of notebooks, $s$, you can buy.

**Answer:** _____ Match? ☐ Yes ☐ No

**8**

$-6 < -2x \leq 4$

**Answer:** _____ Match? ☐ Yes ☐ No